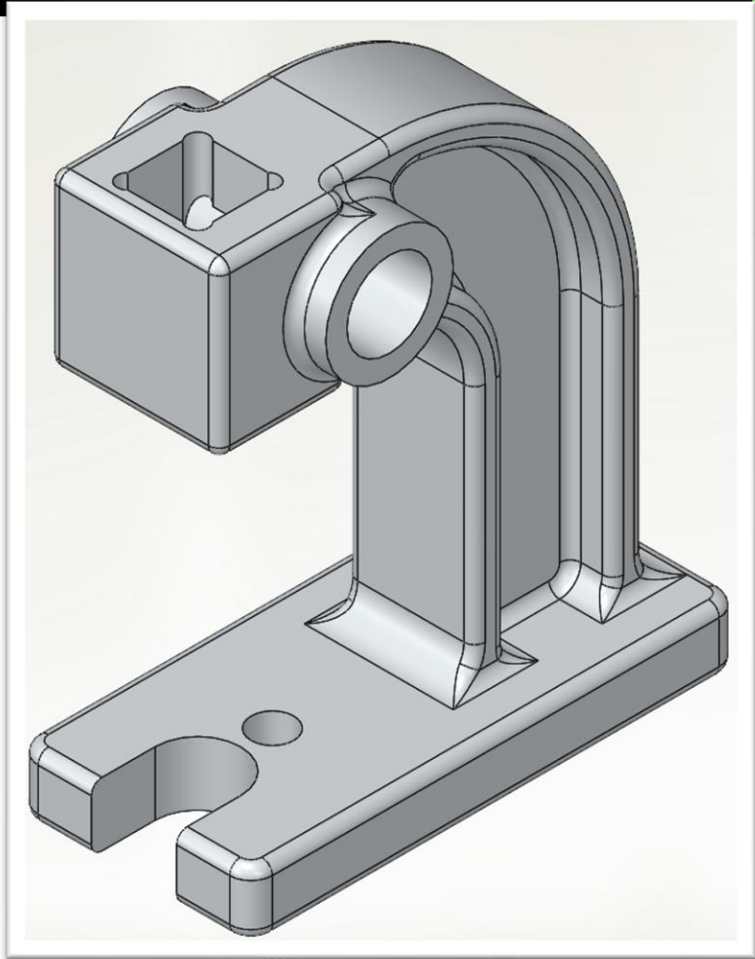


MMAE 445

Arbor Press Design Revision Study



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Introduction

This case study goes through the full computer aided design (CAD) and Mastercam workflow used to model, analyze, and program four variations of an arbor press body, identified as Arbor Press P1 through P4. The study begins with the 3D modeling of each press in SolidWorks, where configurations are used to allow for the ease of changing between design iterations. Static finite element analysis (FEA) is then conducted using a 3 metric ton load to determine stress points and factor of safety characteristics. Two materials are used; 6061-T6 Aluminum for P1 and P3, and Gray Cast Iron for P2 and P4. After FEA, 2D drawings were made for each arbor press. 3D assemblies were then created, including the full arbor press P1 with a rack and pinion mate, a top and bottom cavity mold for arbor press P2, and a core and cavity mold for a shelled version of arbor press P2. The models are then brought into Mastercam to program the first machining operations for arbor press P1 and the arbor press P2 bottom mold, complete with stock setups, roughing, and finishing toolpaths. Once each operation is programmed, it is verified using color loop simulations to ensure accuracy.

Arbor Press P1 CAD & FEA

Prior to modeling Arbor Press P1 through P4, the global document units were set to inches, pounds, and seconds (IPS) and the dimension precision was set to three decimal places (0.XXX).

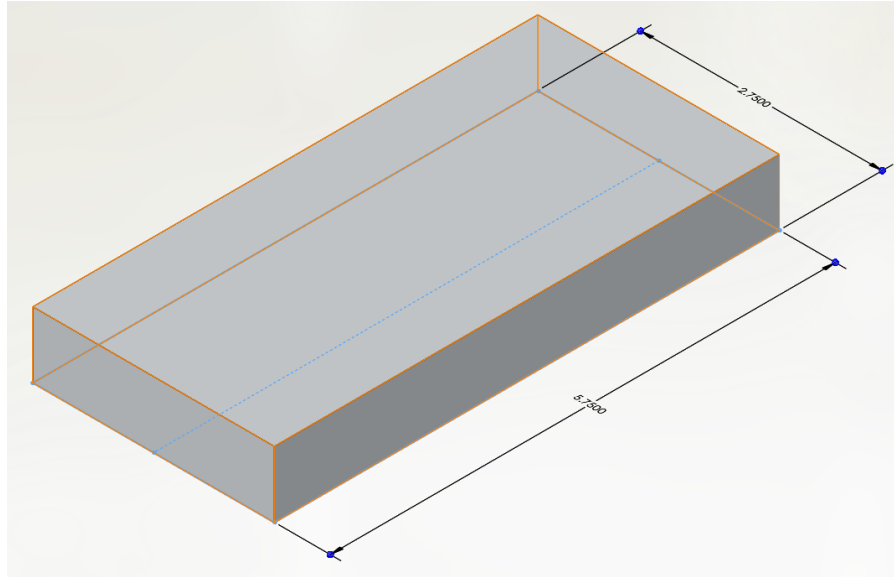


Figure 1: Boss extrude of the base

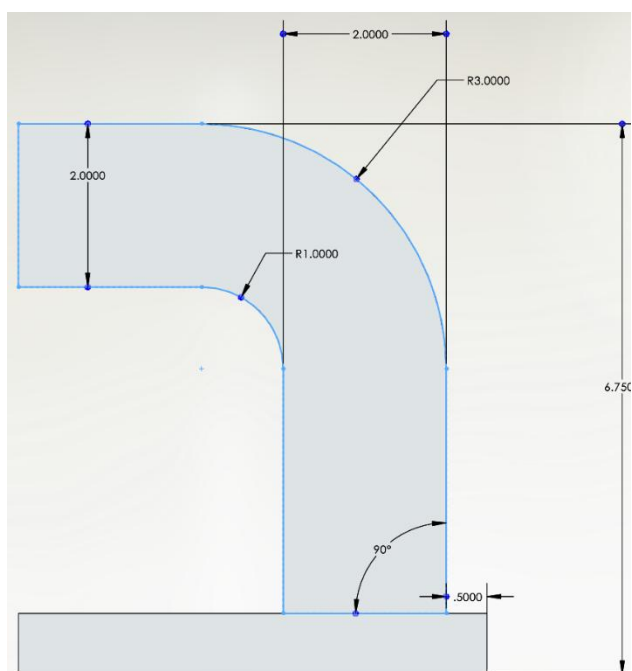


Figure 2: Right plane drawing of arbor press body

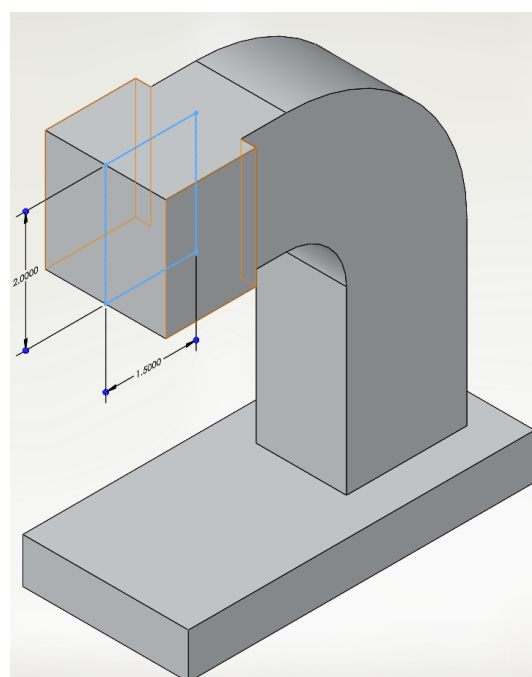


Figure 4: Boss extrude of arbor press head

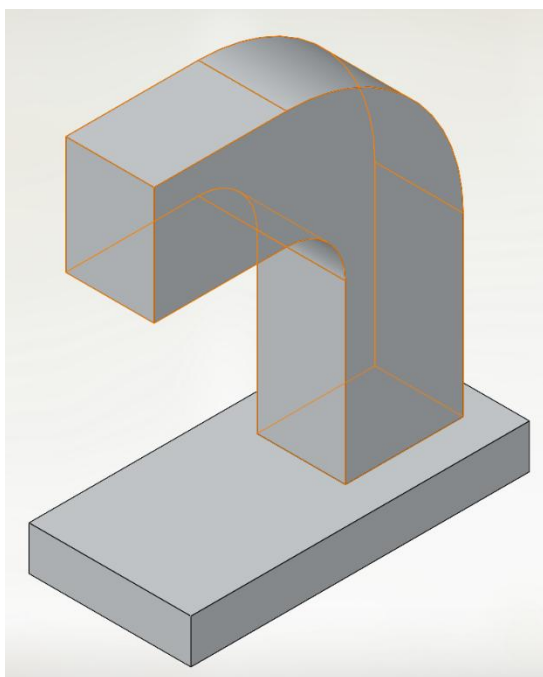


Figure 3: Boss extrude of arbor press body

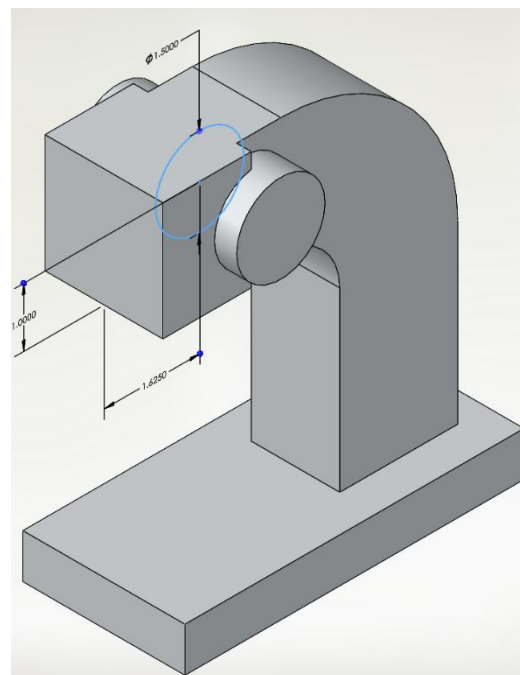


Figure 5: Boss extrude of pinion hole

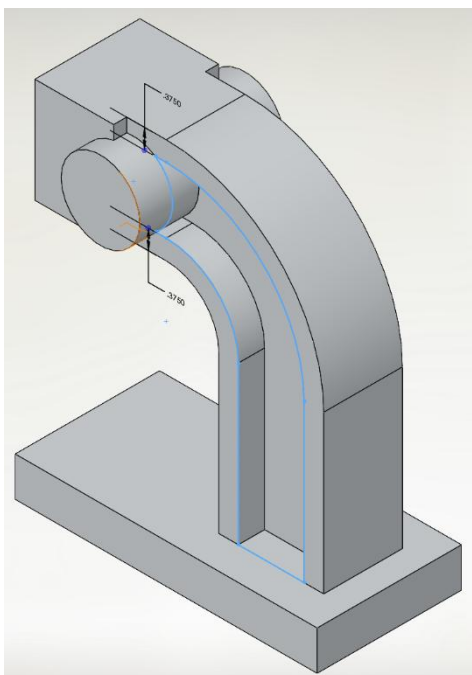


Figure 6: Extrude cut of body pocket

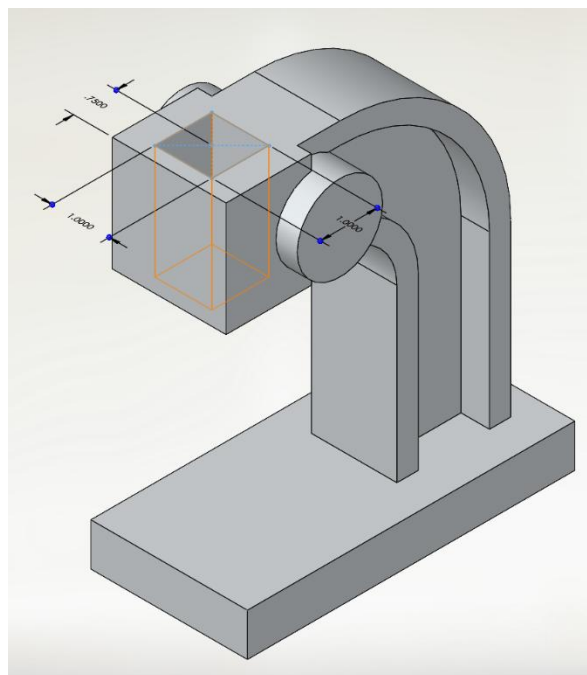


Figure 8: Extrude cut of the ram hole

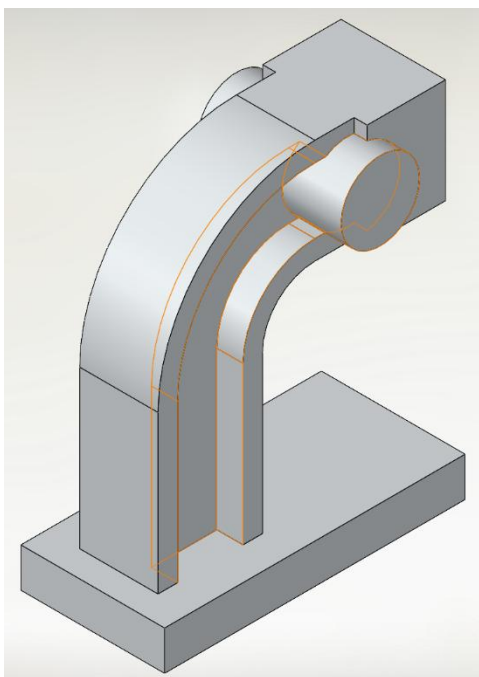


Figure 7: Mirror of the pocket about the right plane

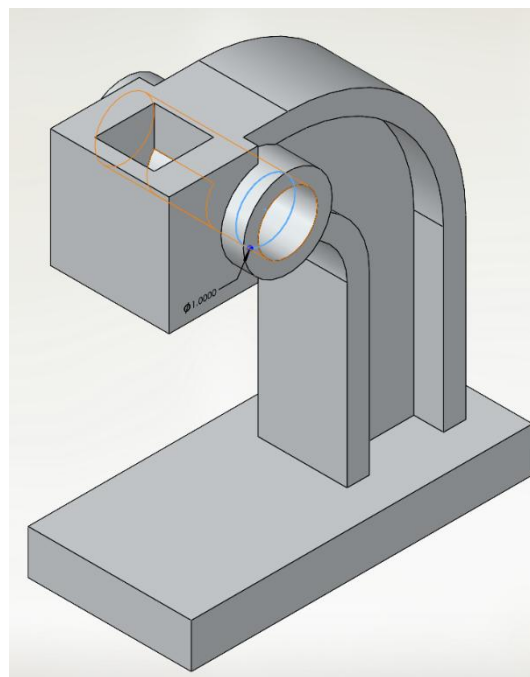


Figure 9: Extrude cut of the pinion hole

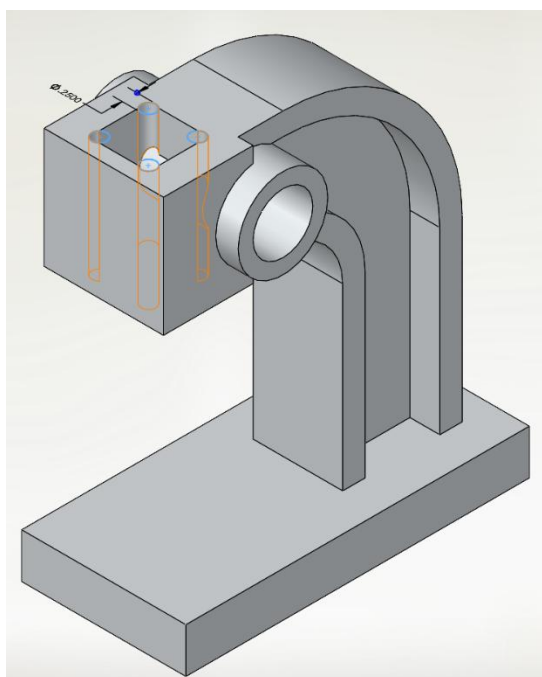


Figure 10: Extrude cut of the ram hole corner reliefs

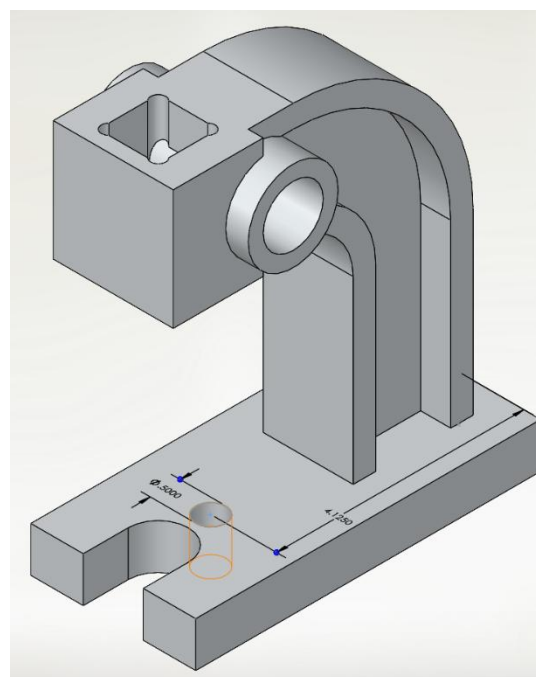


Figure 12: Extrude cut of the anvil hole

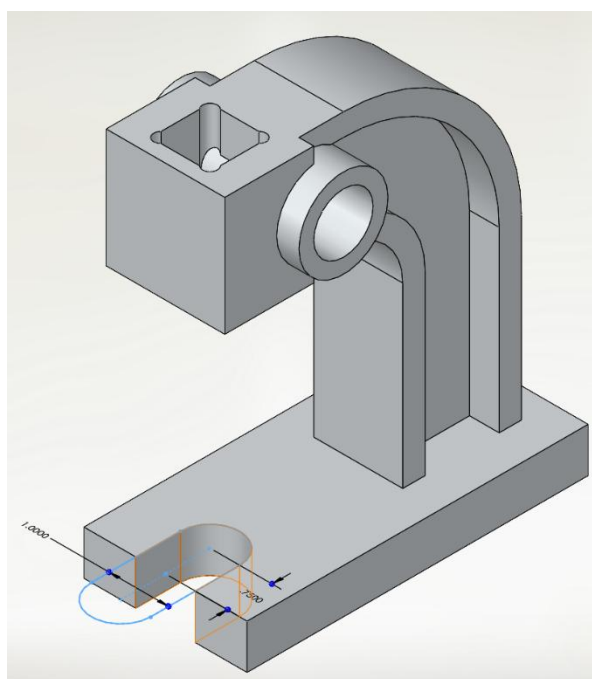


Figure 11: Extrude cut of the arbor slot

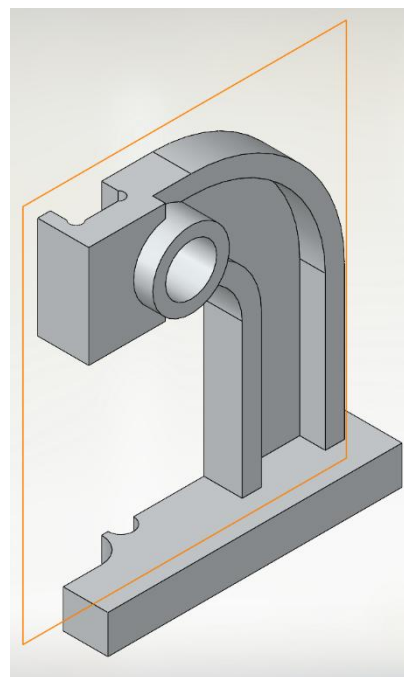


Figure 13: Right plane split and removal of the left half

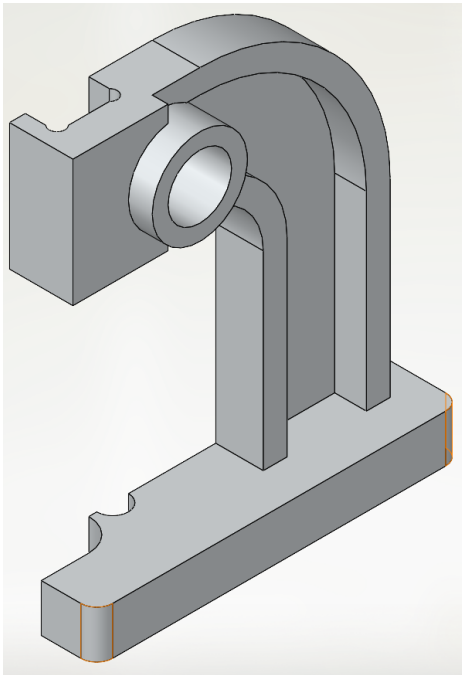


Figure 14: Fillet of the base corners

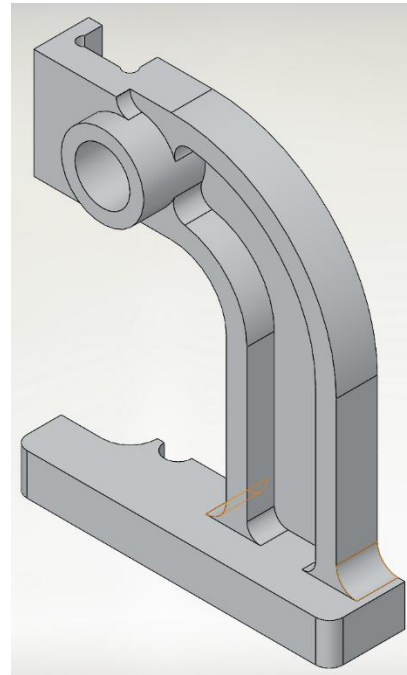


Figure 16: Fillet of the exterior corner radii

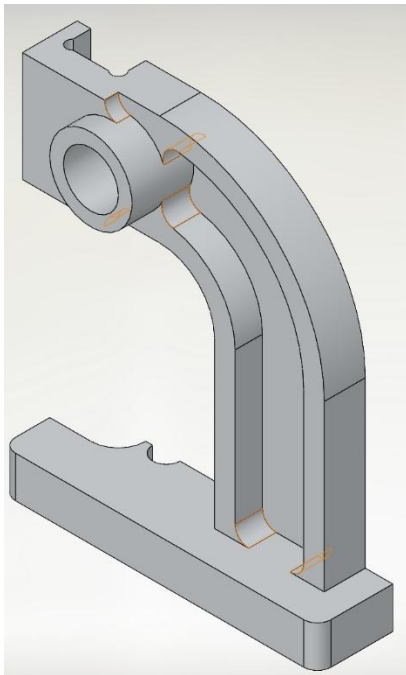


Figure 15: Fillet of the interior corner radii

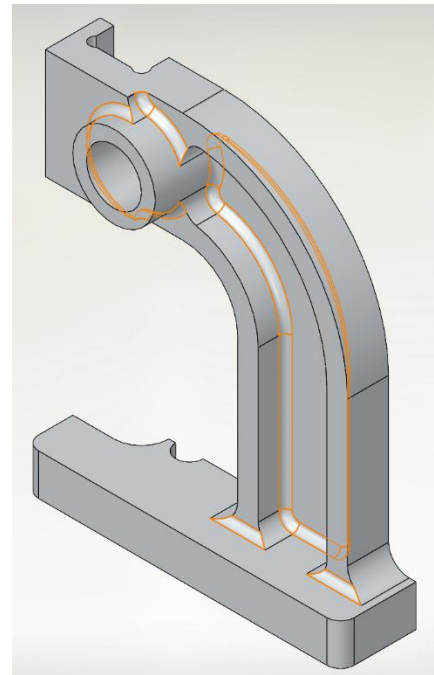


Figure 17: Fillet of the interior floor radii

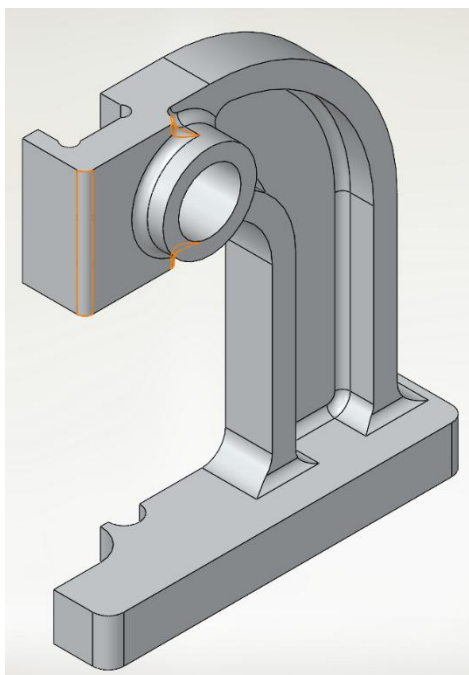


Figure 18: Fillet of the head edges

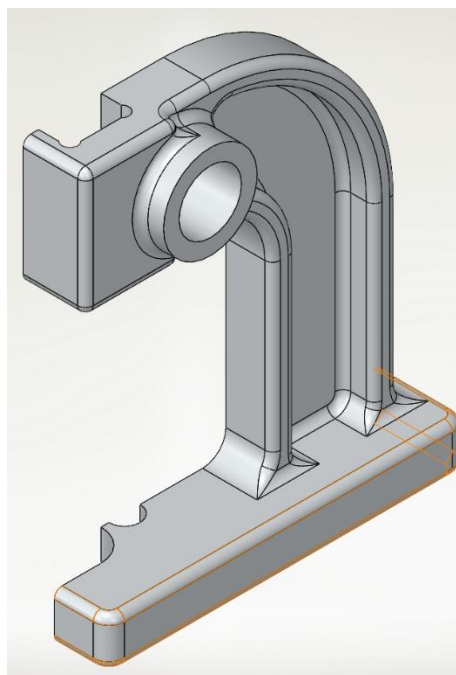


Figure 20: Fillet of the remaining exterior base edges

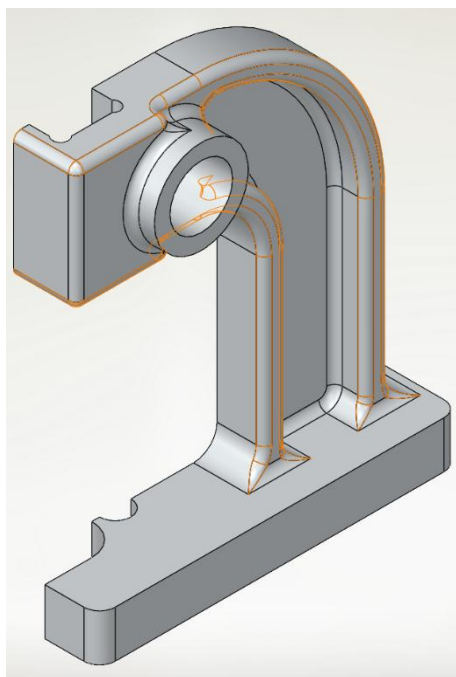


Figure 19: Fillet of the surface radii

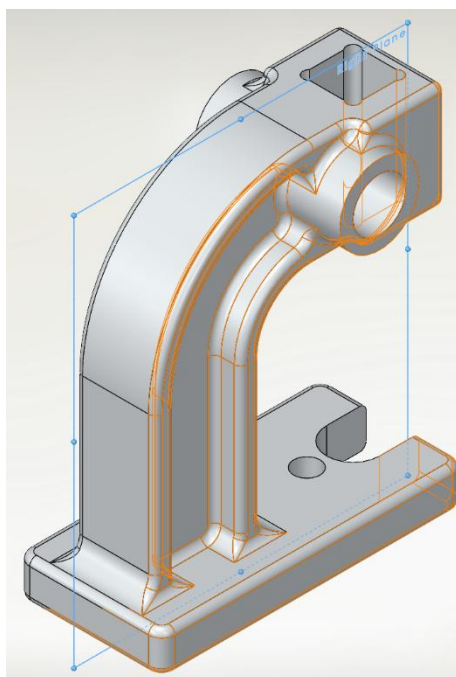


Figure 21: Mirror about the right plane to complete the arbor press

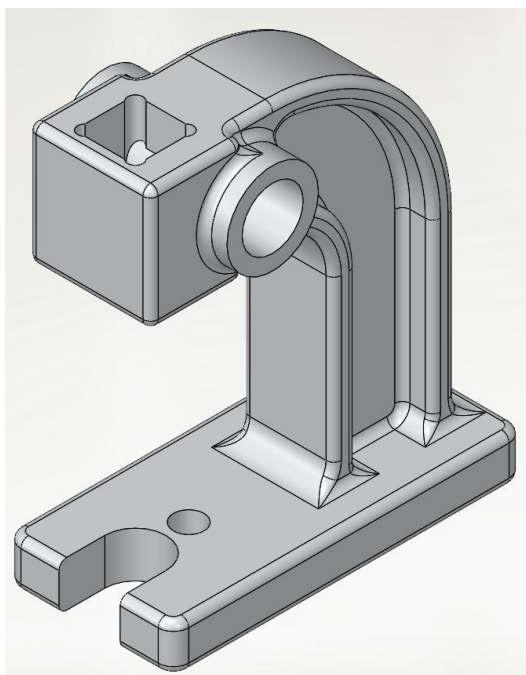


Figure 22: Complete Arbor Press P1

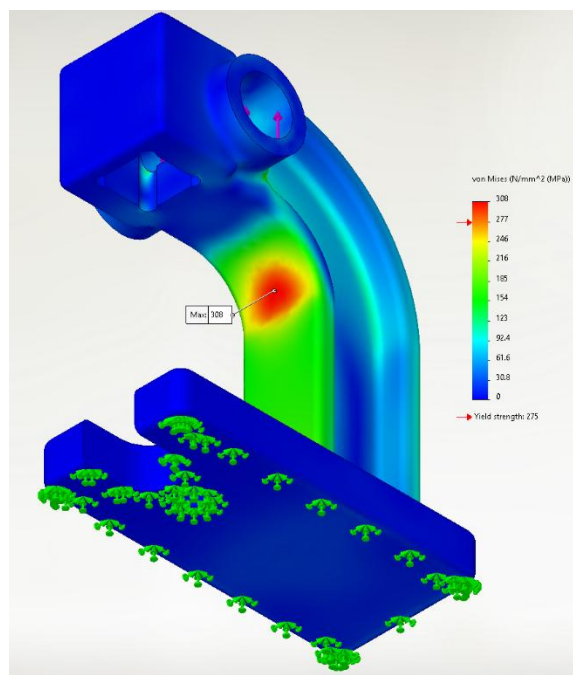


Figure 24: Stress results

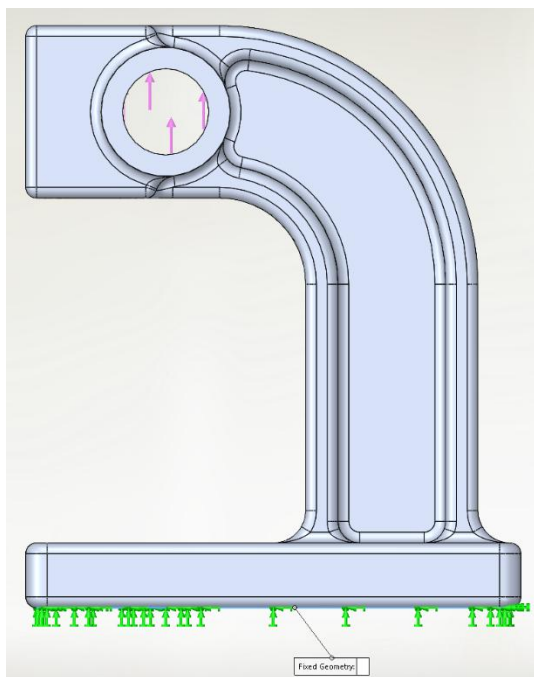


Figure 23: Fixture and Loading conditions for all Arbor presses in FEA

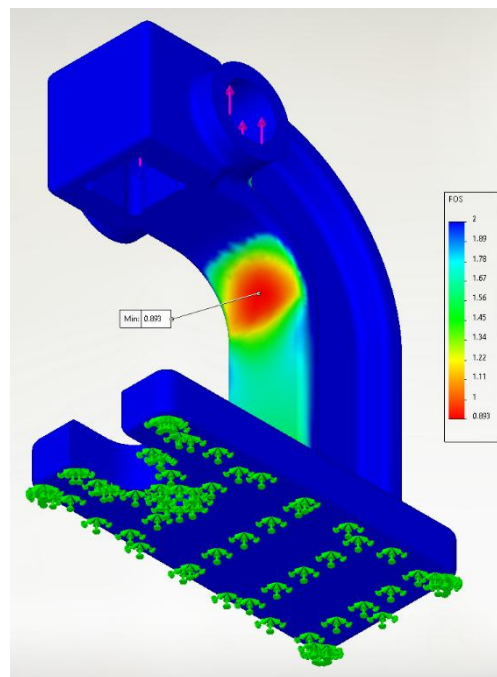


Figure 25: Factor of safety results

Arbor Press P2 CAD & FEA

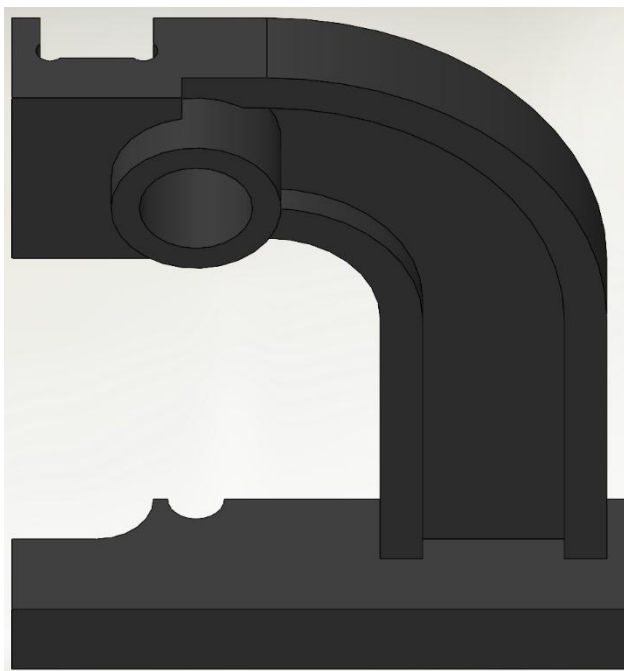


Figure 26: Pre draft application

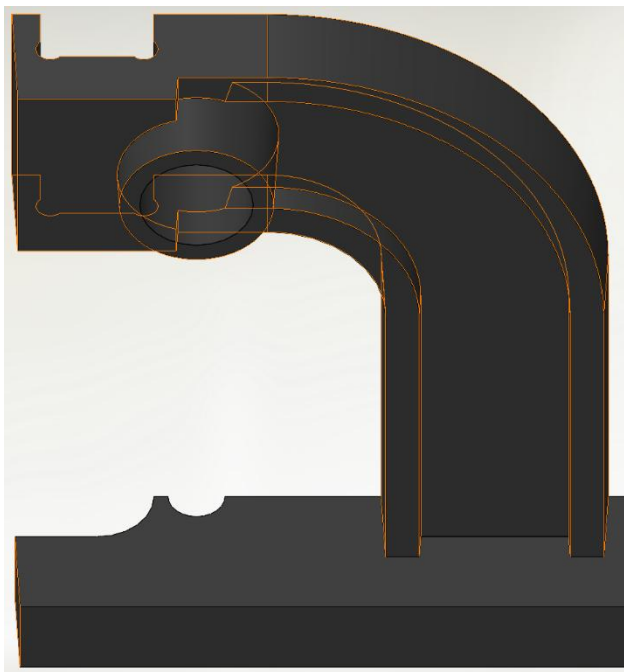


Figure 27: Post draft application



Figure 28: Pre draft application



Figure 29: Post draft application

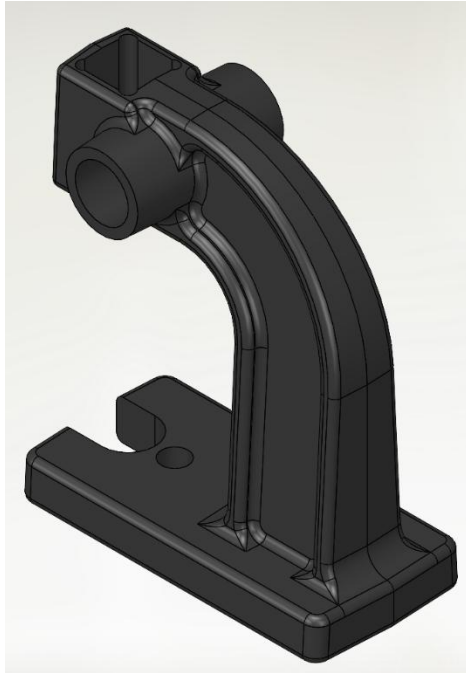


Figure 30: Complete Arbor Press P2

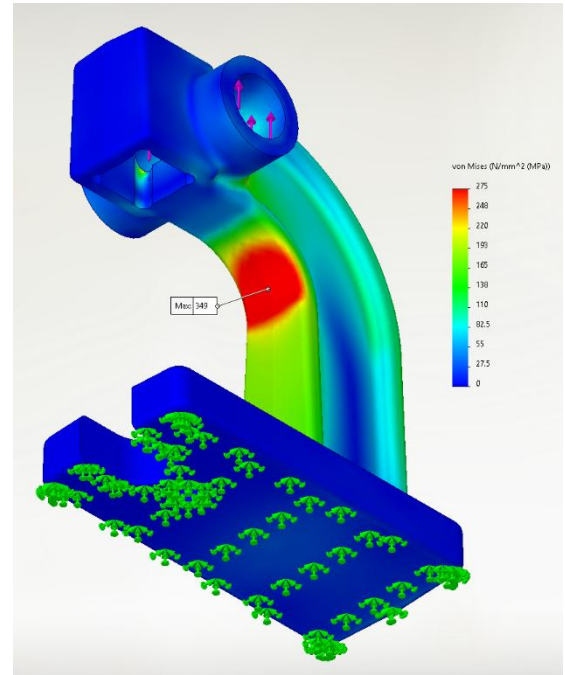


Figure 32: Stress results

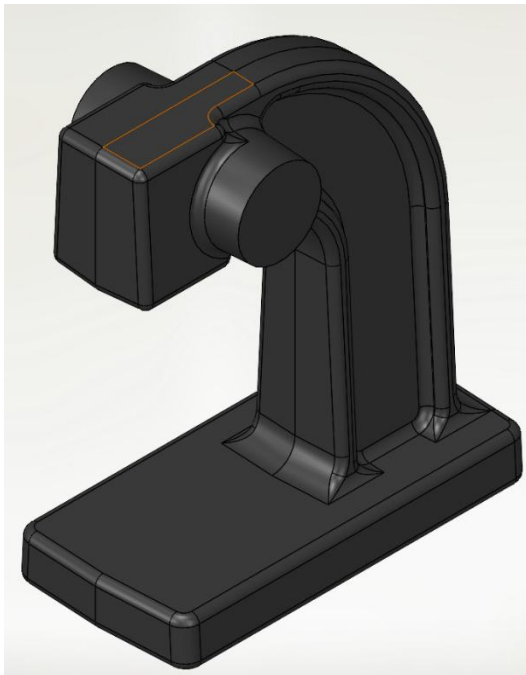


Figure 31: Arbor Press P2 with holes removed for mold making

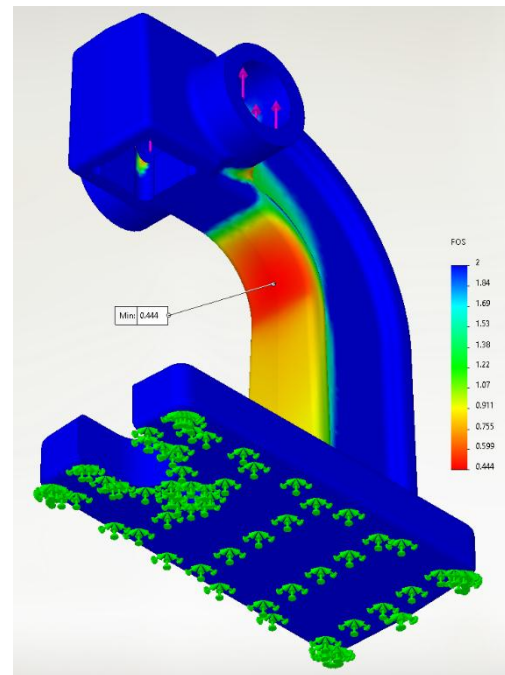


Figure 33: Factor of safety results

Arbor Press P3 CAD & FEA

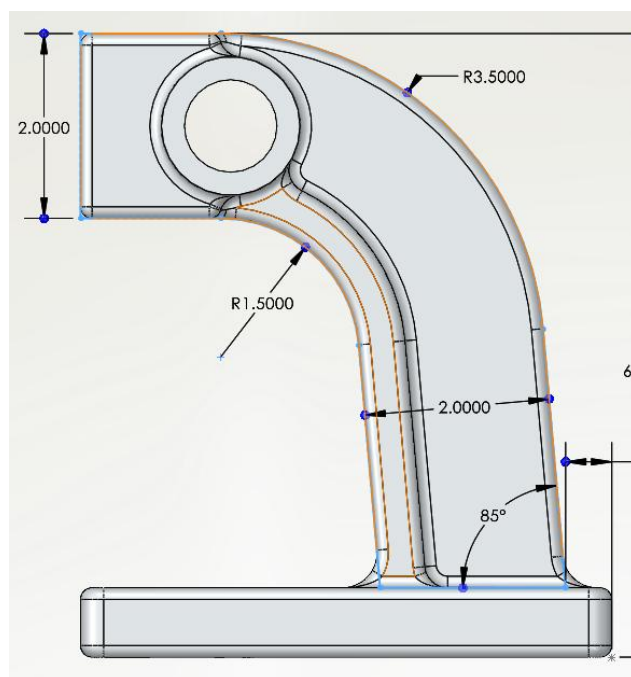


Figure 34: Spine angle change

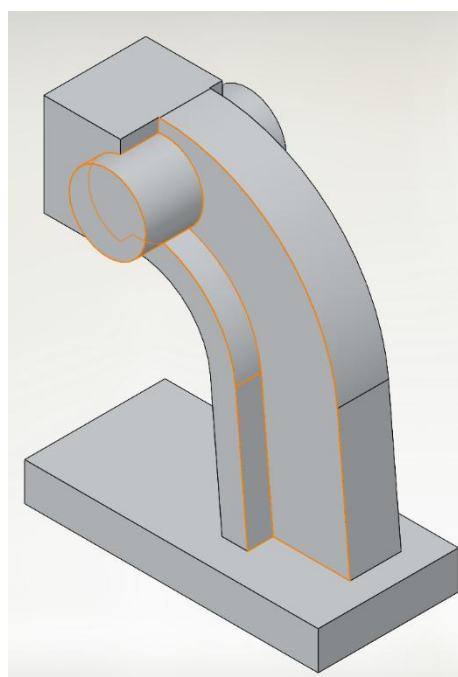


Figure 36: Extrude cut of spine

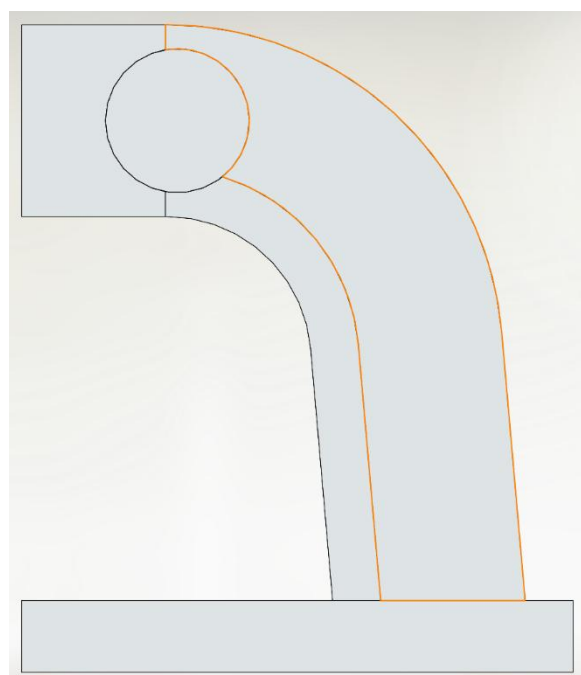


Figure 35: Side pocket right wall removal

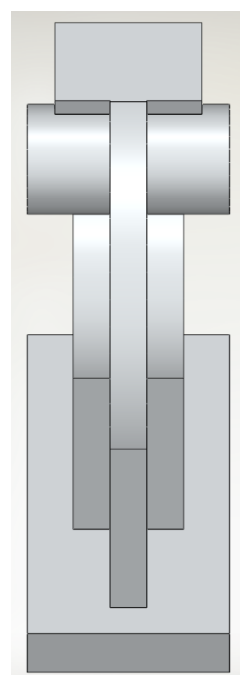


Figure 37: New spine thickness after right plane mirror

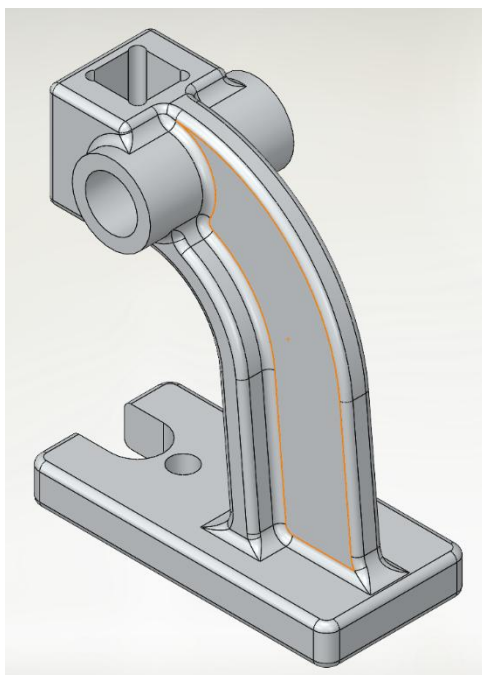


Figure 38: Reapplication of all fillets for final part

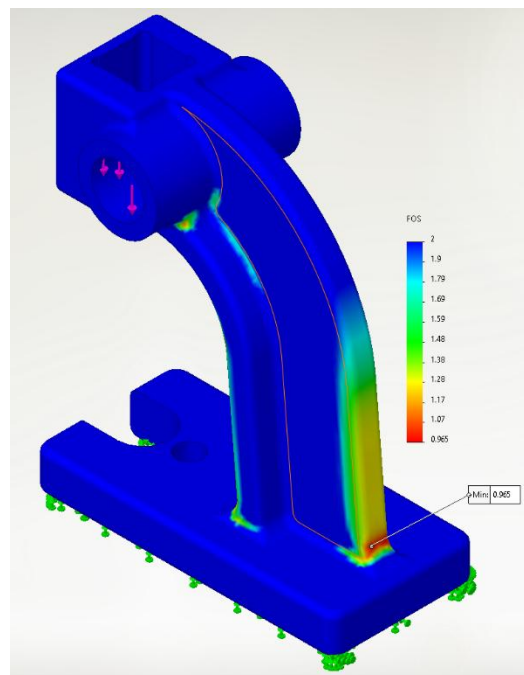


Figure 40: Factor of safety results

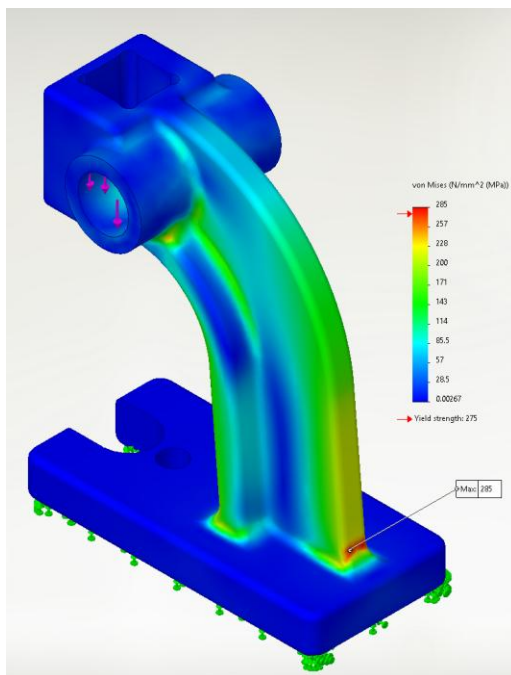


Figure 39: Stress results

Arbor Press P4 CAD & FEA

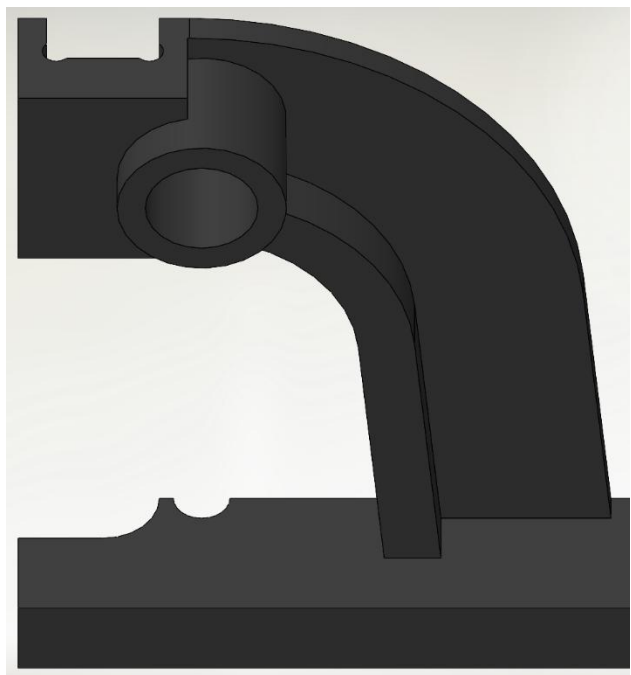


Figure 41: Pre draft application



Figure 43: Pre draft application

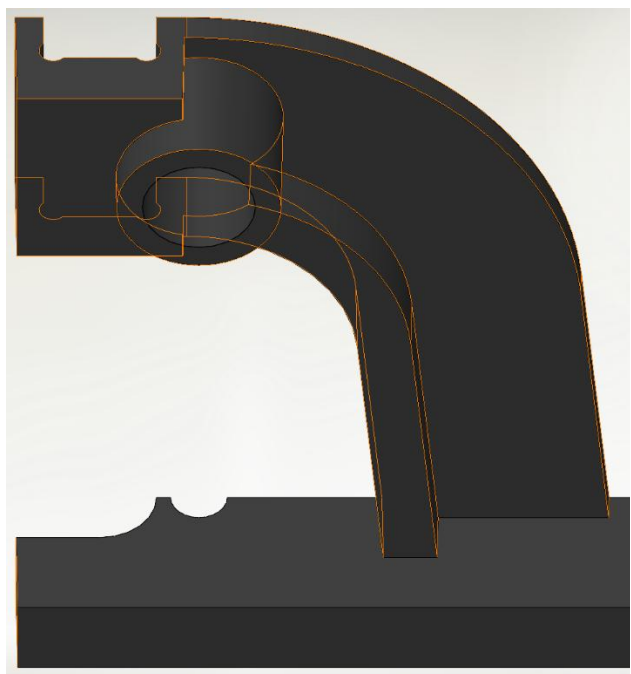


Figure 42: Post draft application



Figure 44: Post draft application

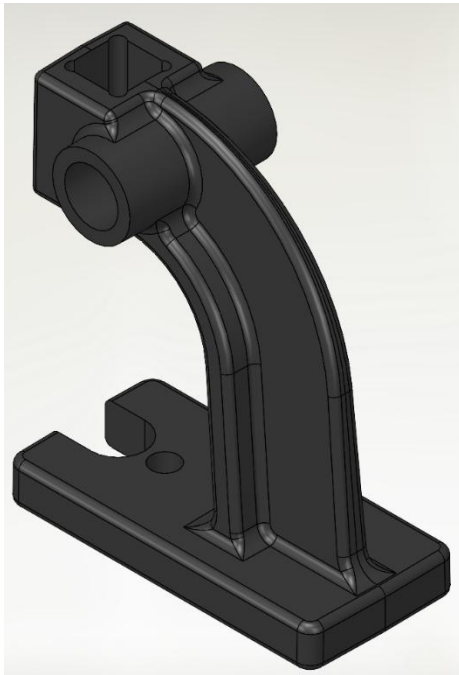


Figure 45: Complete Arbor Press P4

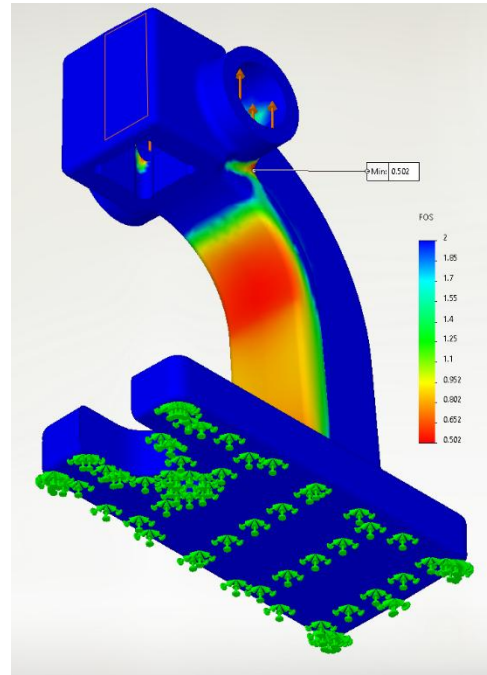


Figure 47: Factor of safety results

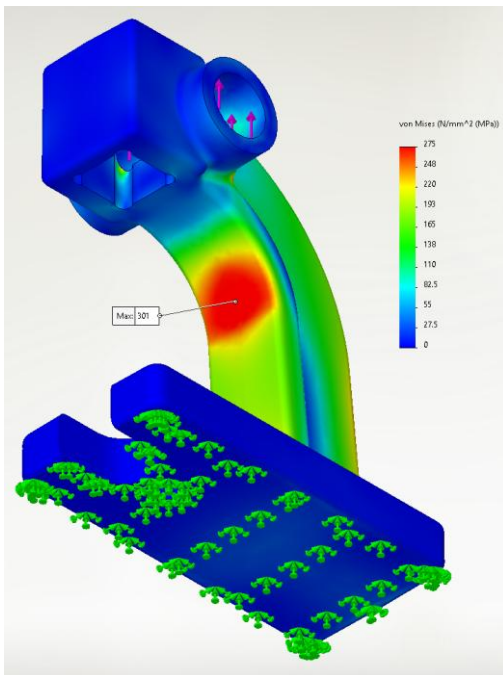


Figure 46: Stress results

Arbor Press P1-P4 Comparison

Table 1: Mass results for each part after completion

Model	Material	Mass (lbs)	Volume (cu in)
P1	6061-T6	6.69	27.57
P2	Cast Iron	6.66	25.6
P3	6061-T6	2.39	24.5
P4	Cast Iron	5.89	22.66

For a fair comparison, Arbor Press P1 and P3 can be compared, while P2 and P4 can be compared. This comes down to the material differences, since aluminum is ductile and cast iron is brittle, meaning that the failure modes is different for each. Aluminum will deform when stressed, giving flexibility on the forces it can take. Cast iron on the other hand will fracture once its yield strength is reached. Looking at Arbor Press P1 and P3, P3 offers a clear mass advantage over P1, with a mass of 2.39 lbs vs 6.69 lbs. For Arbor Press P2 and P4, P4 offers a mass advantage over P2, with a mass of 5.89 lbs vs 6.66 lbs. The general decrease in weight is expected due to the use of drafts on P2 and P4, and the significant removal of material on P3. Of the refined versions, it is expected that the aluminum part is lighter than the cast iron because aluminum is less dense.

Table 2: FEA results for each part

Model	Max Stress (MPa)	FOS
P1	308	0.893
P2	349	0.444
P3	285	0.965
P4	301	0.502

Looking at the FEA results, none of the parts were able to fall below the 275 MPa yield strength, and none of the parts achieved a factor of safety above one, meaning that either the applied loading condition is too high, or further design refinement is needed. P3 was able to come the closest with a maximum stress of 285 MPa and a near 1 FOS, but was unable to cross either threshold. For future design iterations, the spine of the part can be made thicker with less curvature along the outside edge to add rigidity.

Arbor Press P1-P4 & Assembly Drawings & GD&T

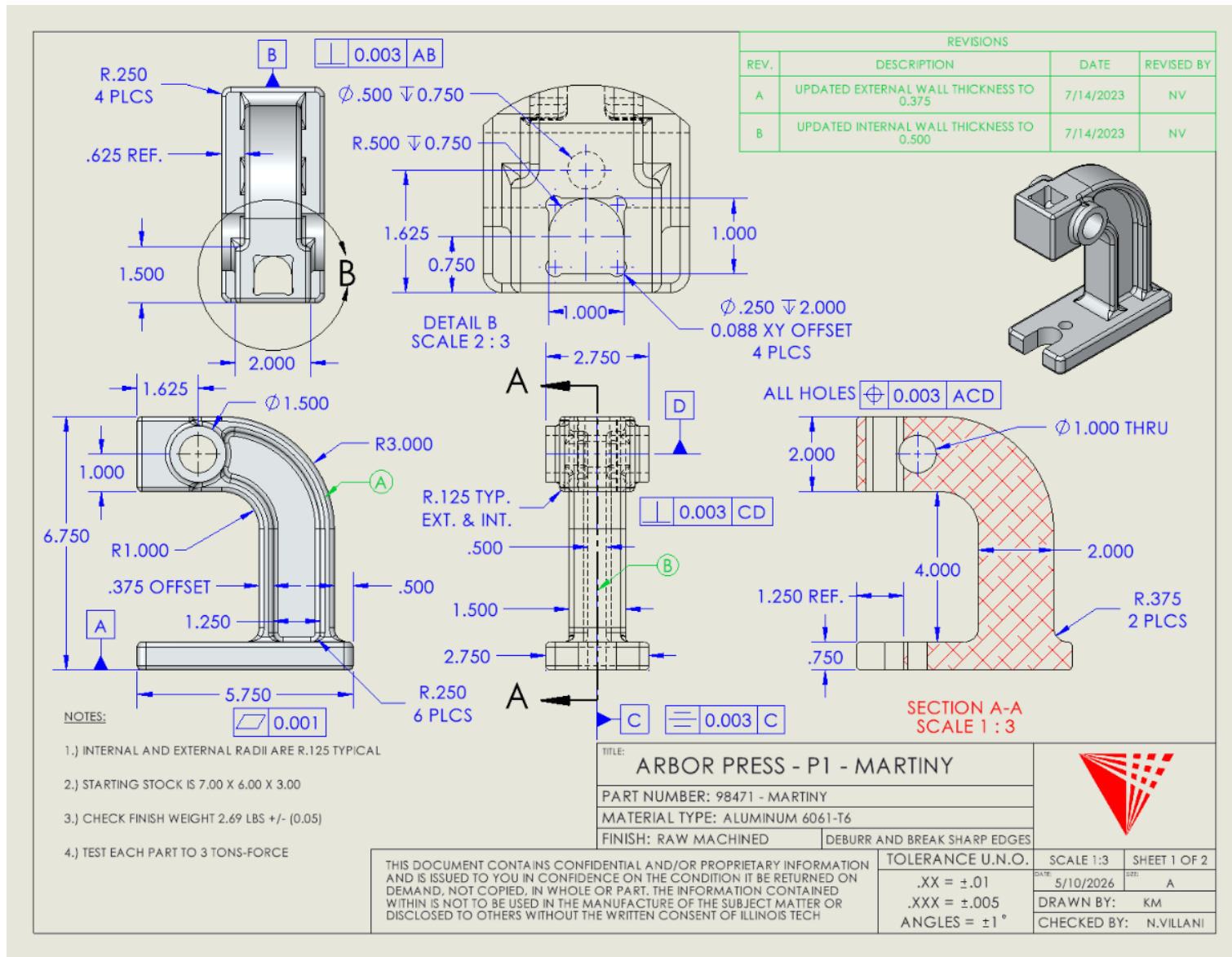


Figure 48: Arbor Press P1 reference drawing

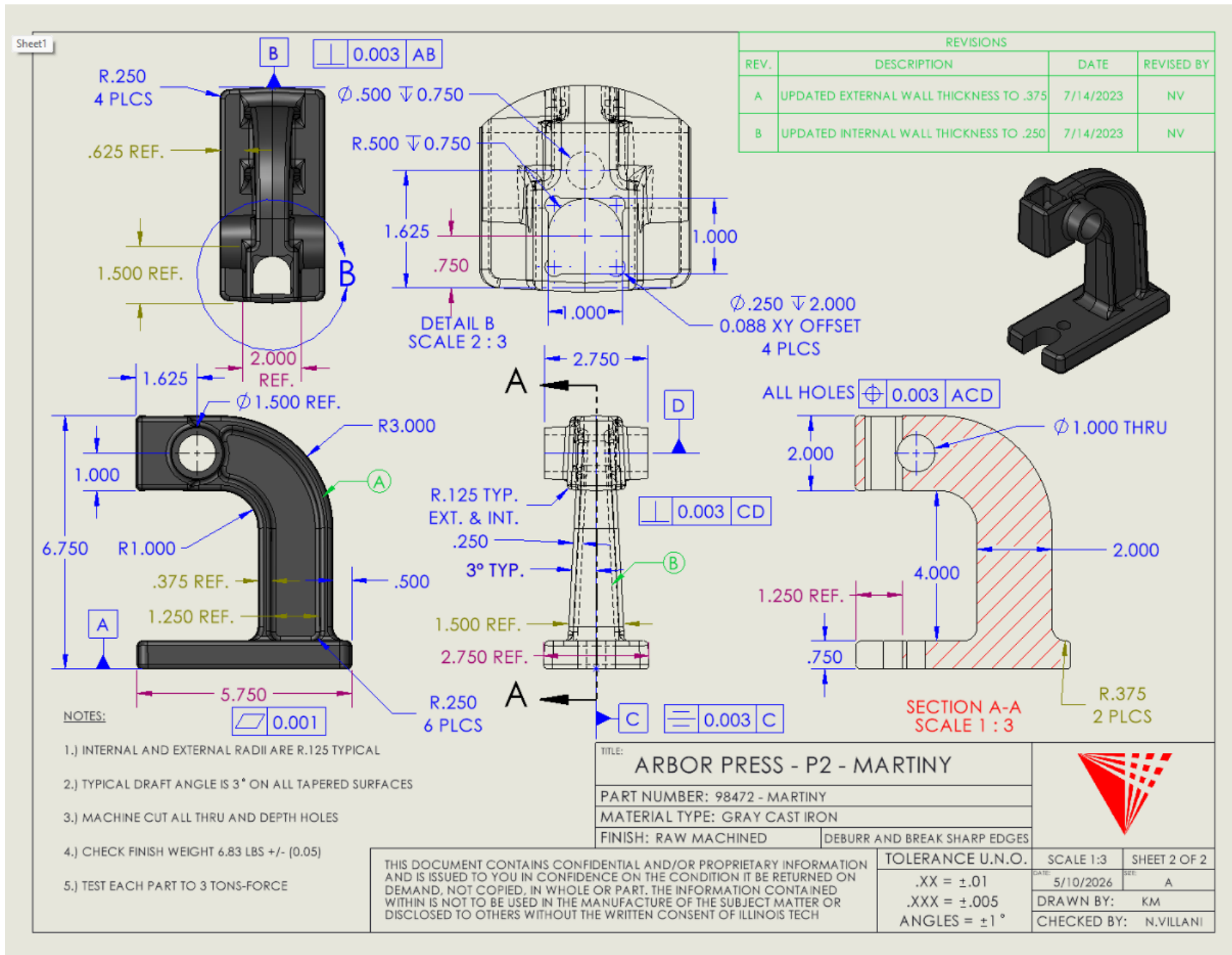


Figure 49: Arbor Press P2 reference drawing

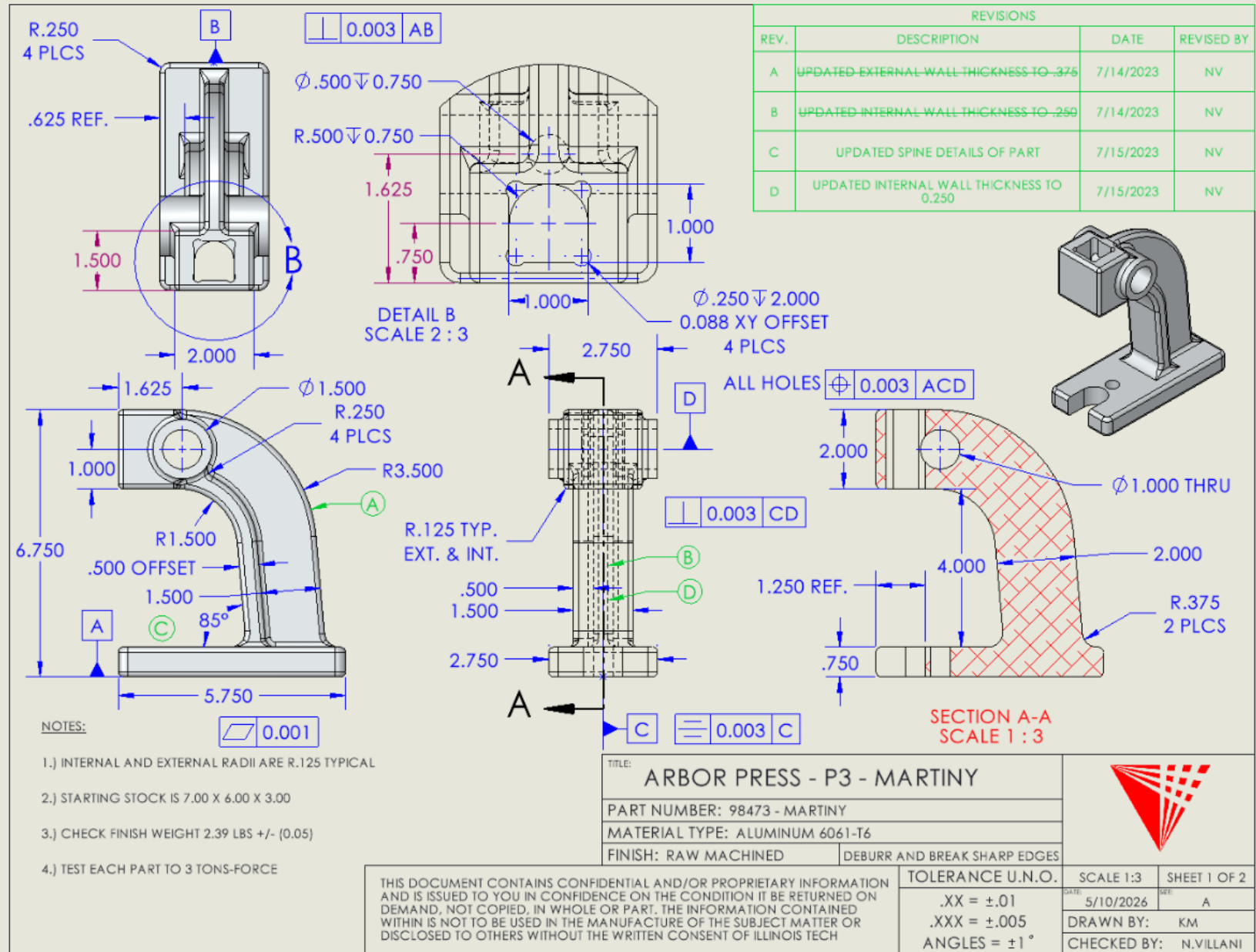


Figure 50: Arbor Press P3 reference drawing

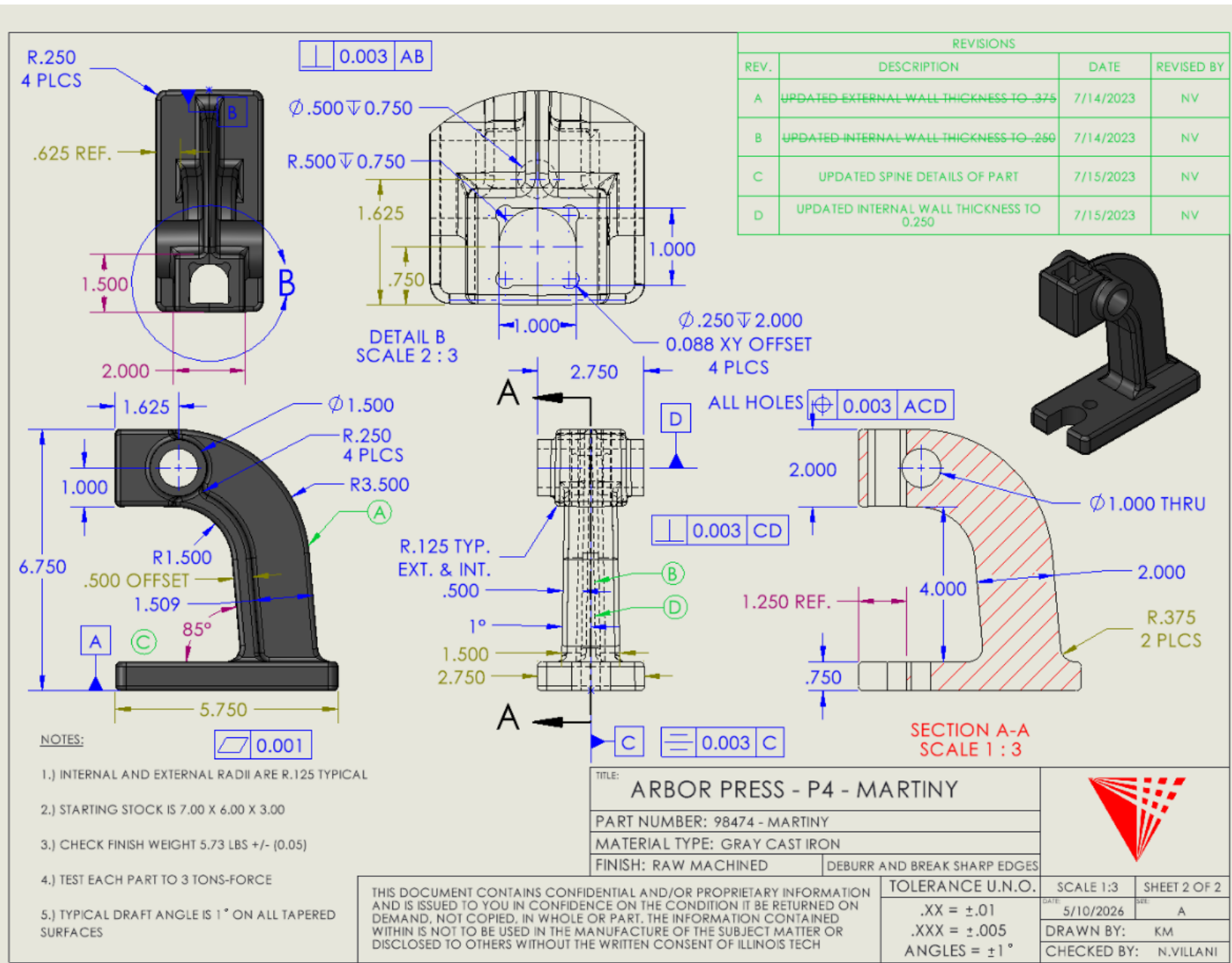


Figure 51: Arbor Press P4 custom drawing

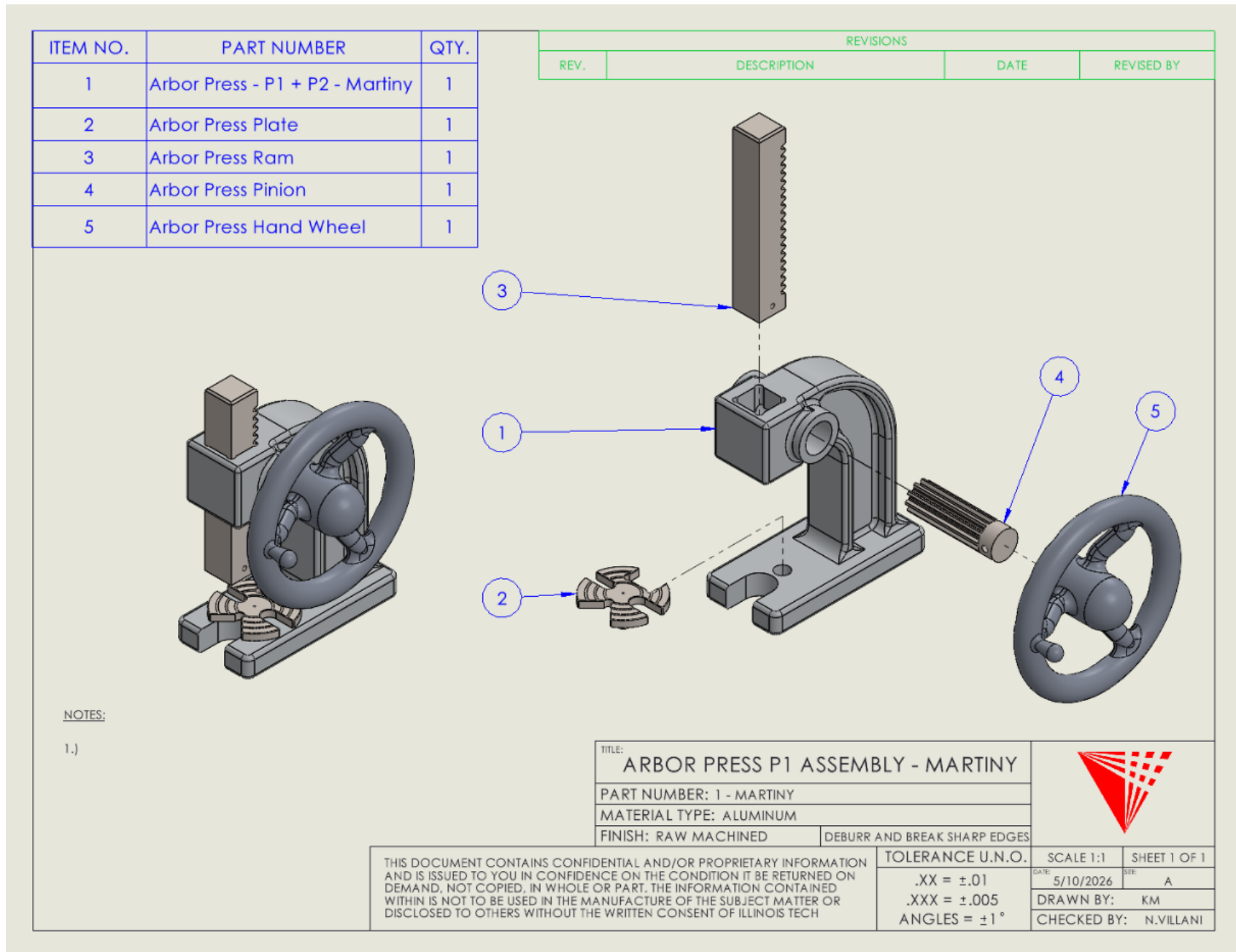


Figure S2: Arbor Press P1 assembly exploded view and bill of materials

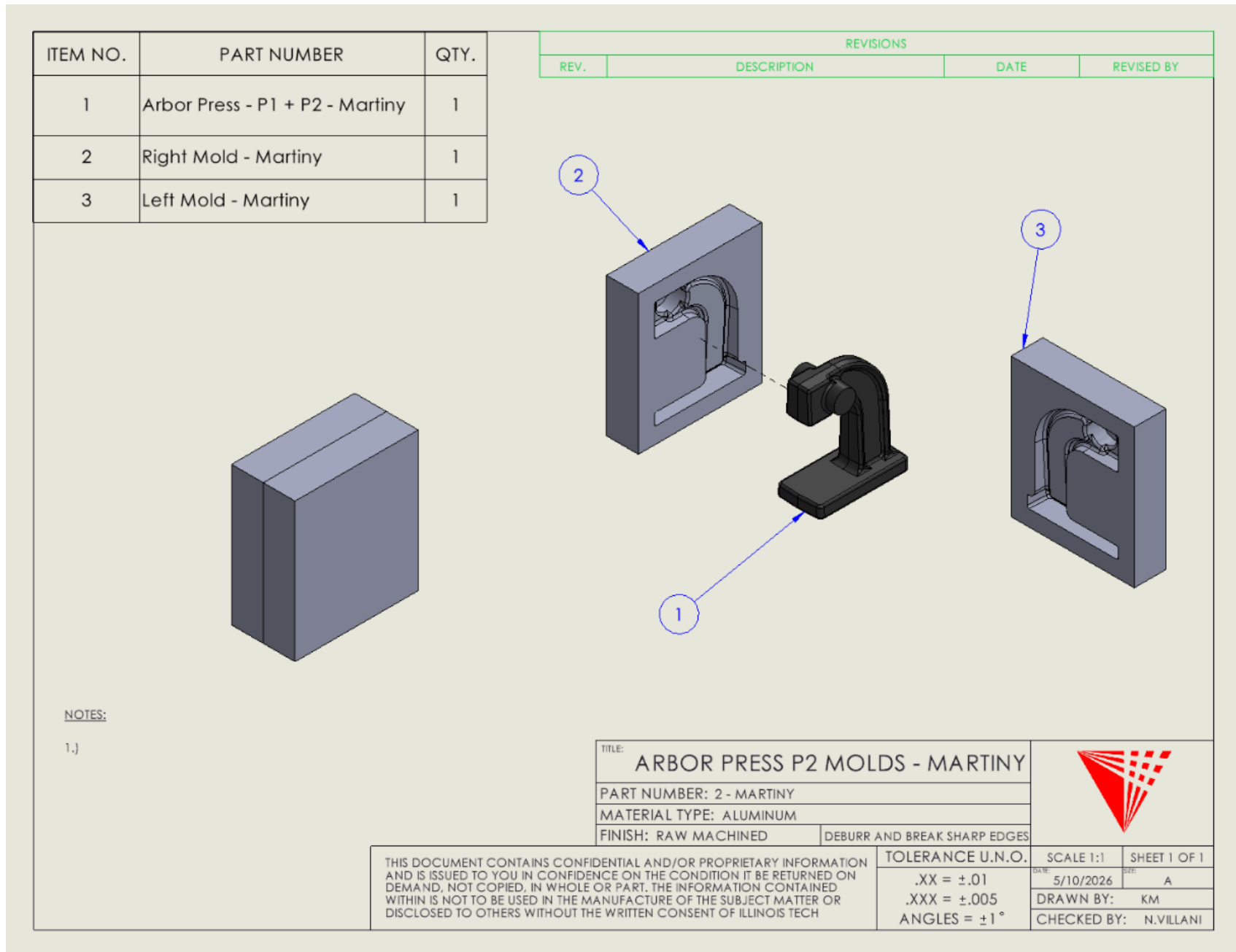


Figure 53: Arbor Press P2 left and right molds exploded view and bill of materials

There are five views that are used for the drawings in Figures 48 to 53. They are a front orthographic view, a section view, a detail view, a shaded isometric view, and an exploded view. The front orthographic view gives the overall shape of the part and gives a clear picture of the major geometric details like the mounting holes and wall thickness. Section A-A is used to give an internal picture of both parts and to allow for the dimensioning of internal features like floor thickness, stepped geometry, and internal radii. These features would not be clearly visible in other exterior views. Detail B is used to enlarge small details like the corner radius and small hole dimensions. These dimensions could be added to other views but would result in crowding. The shaded isometric view gives an overall impression of how the part is supposed to look when it's done. It is not used for any dimensioning or tolerancing but is useful for manufacturing. The exploded view is used for assembly drawings to show how parts fit together, and is aided by the bill of materials.

Symbol	Name	Type	Description	Active Drawing
	Position	Location	A position details the location of a hole, slot, or circular feature	P1-P4
	Perpendicularity	Orientation	A 90-degree orientation between features	P1-P4
	Flatness	Form	Flatness states that an entire surface lies in one plane	P1-P4
	Symmetry	Location	Symmetry establishes a center plane and makes points on each side of the plane equal	P1-P4

Each of the geometric dimensioning and tolerancing (GD&T) symbols used in Arbor Press P1-P4 drawings plays an important role in determining how the part will function after it has been manufactured. Position is used to declare the exact position of the holes because their location determines how the final part works with other parts. The holes might have the correct diameter, but if their centers are not correctly placed, there could be assembly issues after manufacturing.

Profile of a line is used on the arcs because it is important for the inner and outer curves to maintain their shape. Perpendicularity is used to keep a feature at a square orientation relative to the chosen datums. Flatness is used to keep a surface planar and makes sure that the contact between the part and another surface or part is even, with very little warping. Symmetry is used heavily in all four parts here as the part was designed around the midplane, meaning that across the midplane, every point needs to be identical. Having symmetry ensures the part is balanced and loads are distributed evenly. It also allows for simplified modeling, since only one half needs to be done at a time.

The GD&T tolerances are different from the tolerances called out in the title block because they control specific geometric requirements that a general size tolerance cannot. The title block tolerance gives a default, general value for the allowed variation in dimensions, but does not control the specifics of a feature that might be required for proper function. GD&T is used on performance critical features where the function, fit, and manufacturability matter more than the general sizing alone. For all press drawings, the tighter tolerances make sure that the holes align correctly and work with other parts in the assembly and important surface features remain properly oriented relative to each other and defined datums.

Arbor Press P1 Programming



Figure 54: Tool paths used for the top half of Arbor Press P1

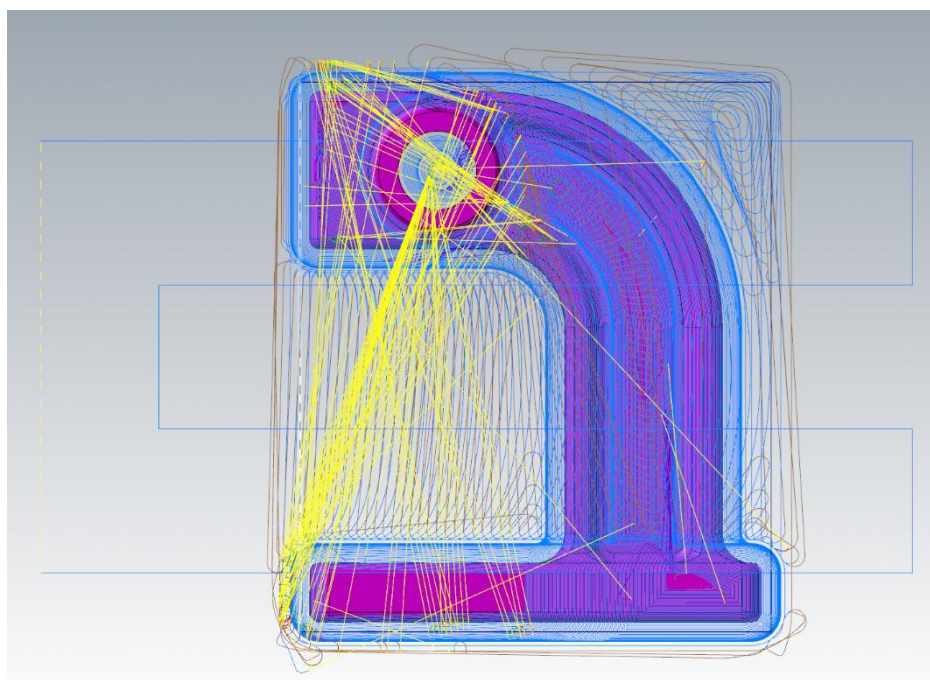


Figure 55: Top view of all tool paths

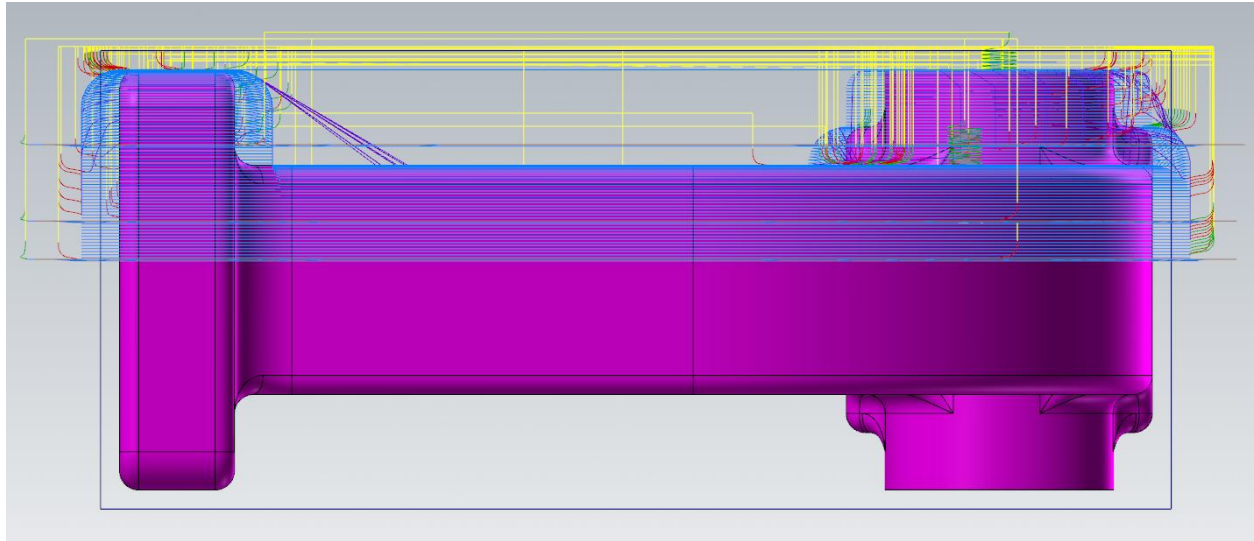


Figure 56: Side view of all tool paths

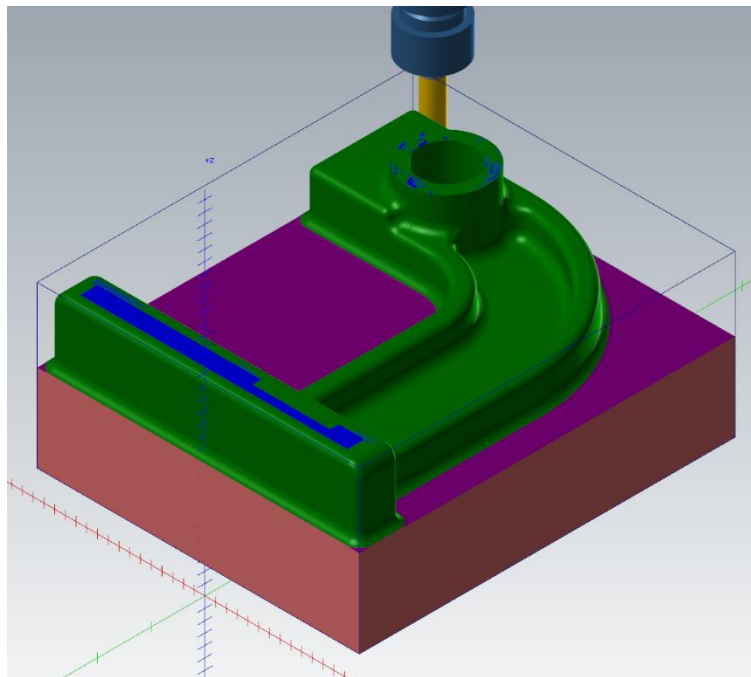


Figure 57: First operation verification using color loop

Arbor Press P2 Programming



Figure 58: Tool paths used for first operation of Arbor Press P2 right mold

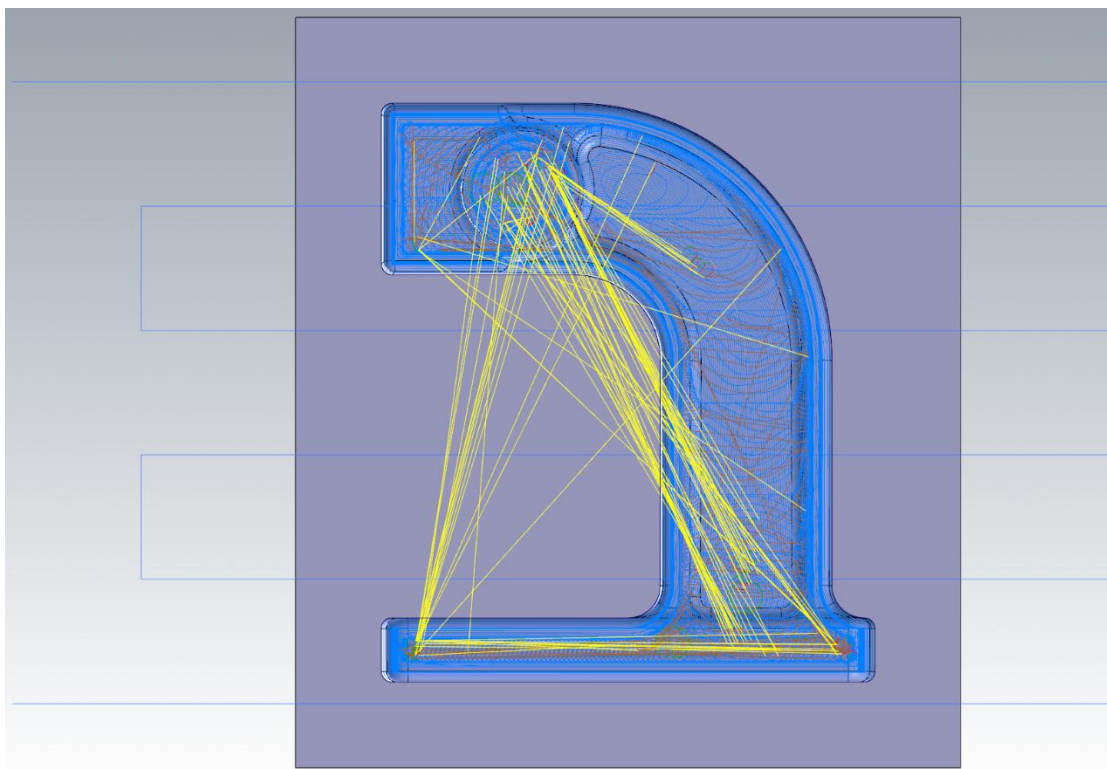


Figure 59: Top view of all tool paths

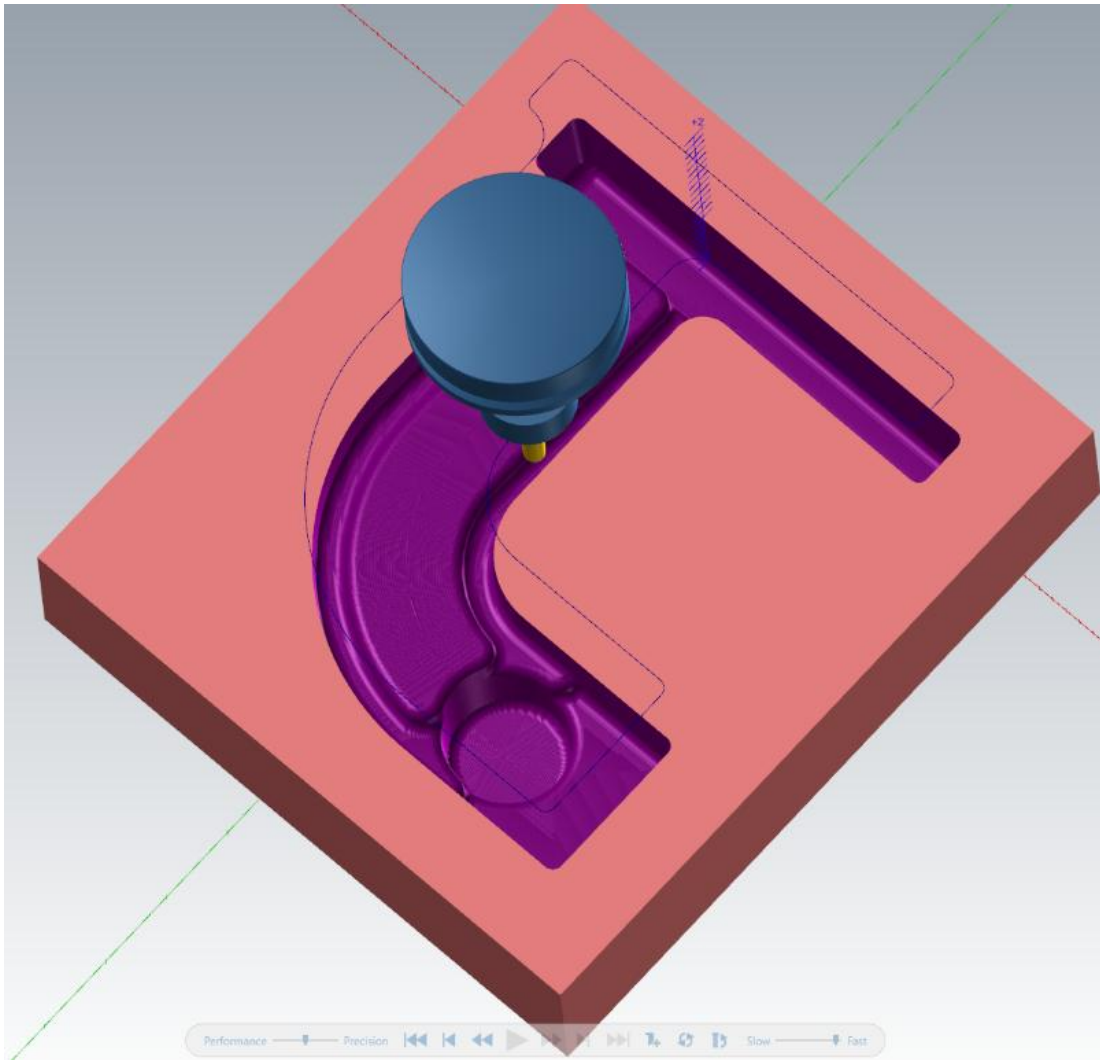


Figure 60: First operation verification using color loop

The reason for making a mold is to allow a molten metal or plastic to be poured or injected to quickly and repeatedly make a part. In order for the part to be removed from the mold, all interior features need to be removed, so that once the material cools, it does not lock the part into the mold. Looking at the through hole on the head of the arbor press in Figure 22, if the part is cast in the position from Figure 60, material will flow through that hole and solidify in the space. Once everything cools, that hole would lock the part in place because it is perpendicular to the direction of separation. Filling in all the holes ensures that the only parts in contact with the mold are on the outer surface.

Arbor Press P4 Plus CAD & FEA

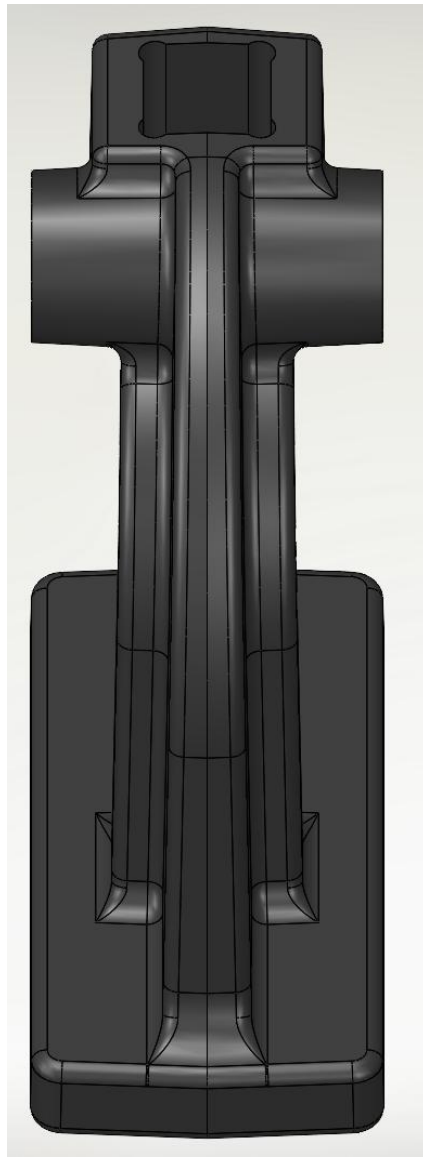


Figure 61: Further P4 modification to thicken the spine

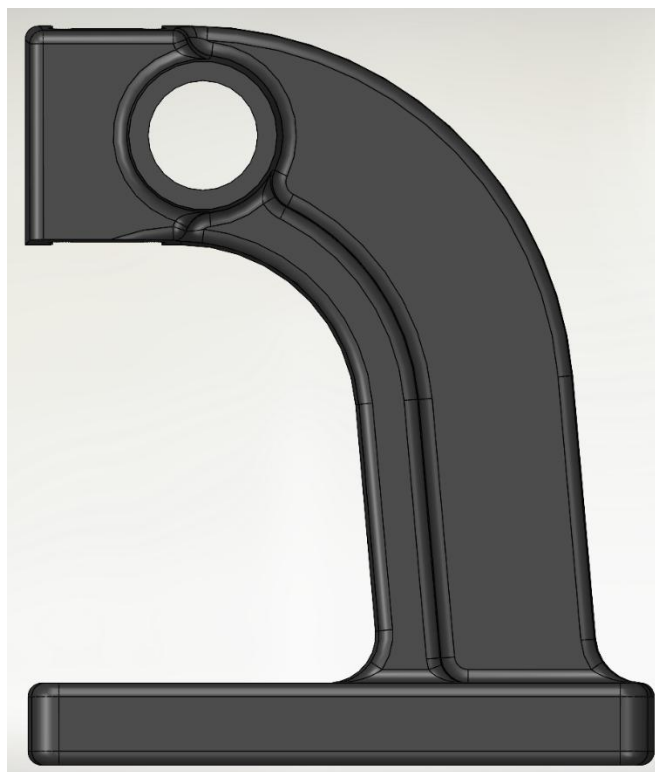


Figure 62: Further P4 modification to thicken the front spine wall

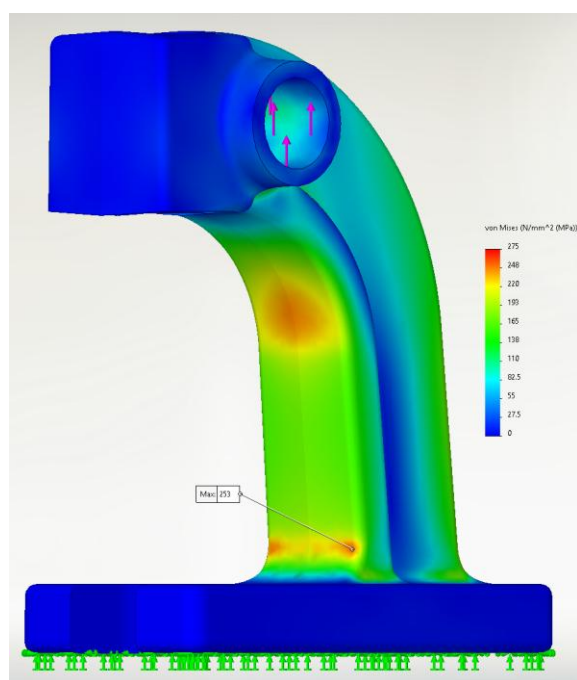


Figure 63: Stress results

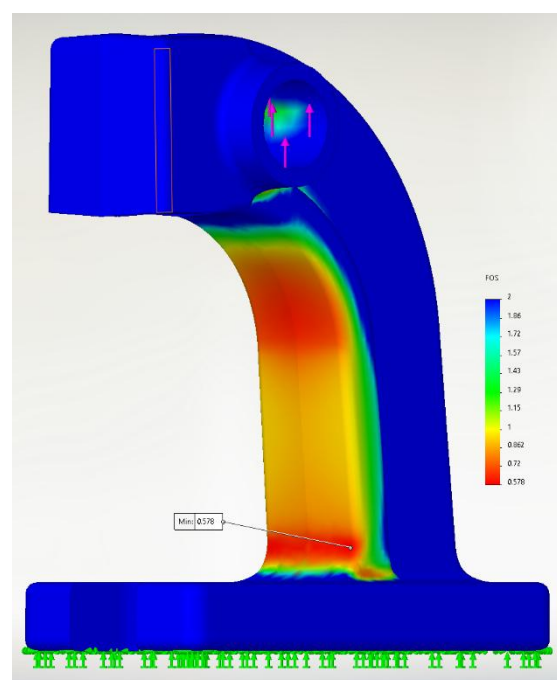


Figure 64: Factor of safety results

In addition to the four Arbor Presses already made, I decided to make another to try to lower the max stress and increase the FOS. By thickening the spine using a 1 degree draft and increasing the wall thickness, a maximum stress of 253 MPa was achieved with a FOS of 0.578. This was done with the part weighing 6.34 lbs. The inability to reach a higher FOS primarily comes down to the material choice. As mentioned above, cast iron is brittle, which means that in order to have a high FOS, the part needs to be over engineered. Switching the material from cast iron to aluminum will increase the FOS.

Extra Credit

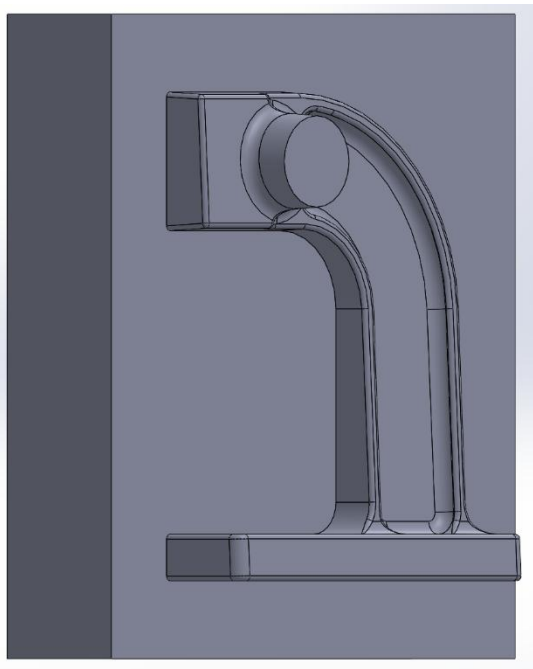


Figure 65: Arbor Press P2 core mold

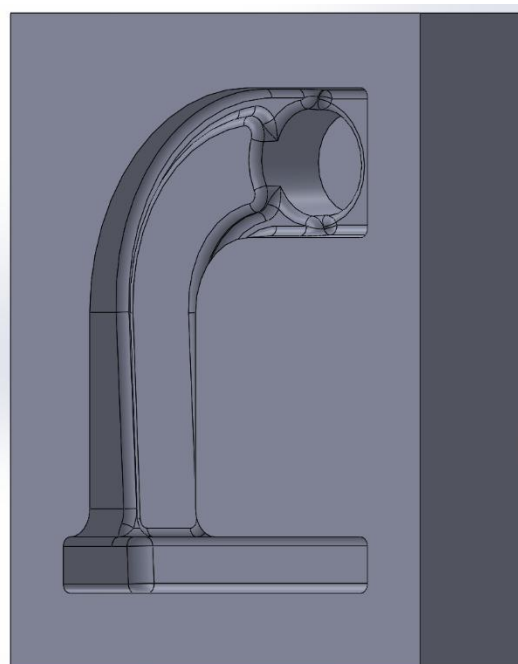


Figure 66: Arbor Press P2 cavity mold



Figure 67: Tool paths used for core mold

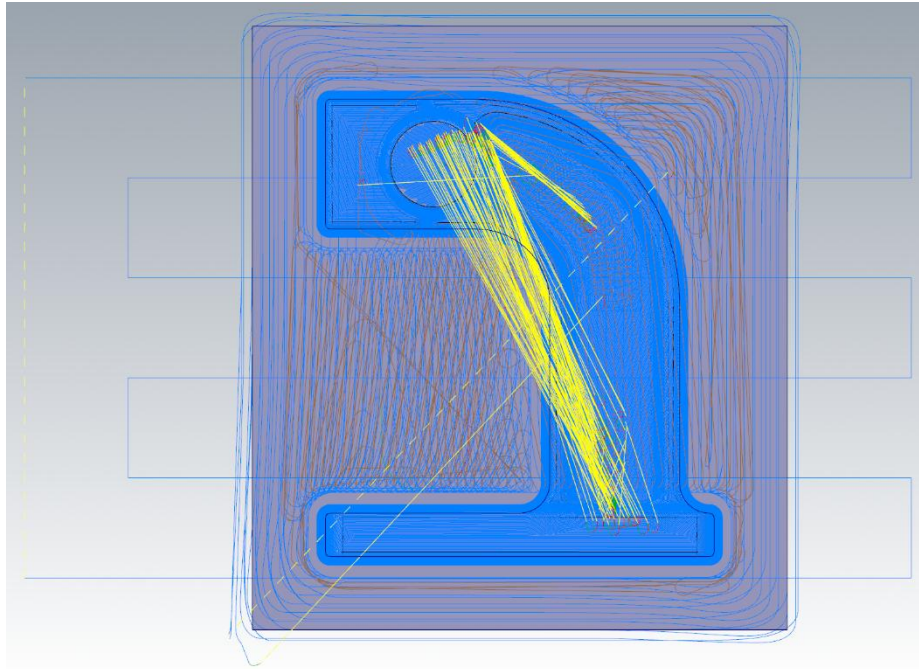


Figure 68: Top view of tool paths

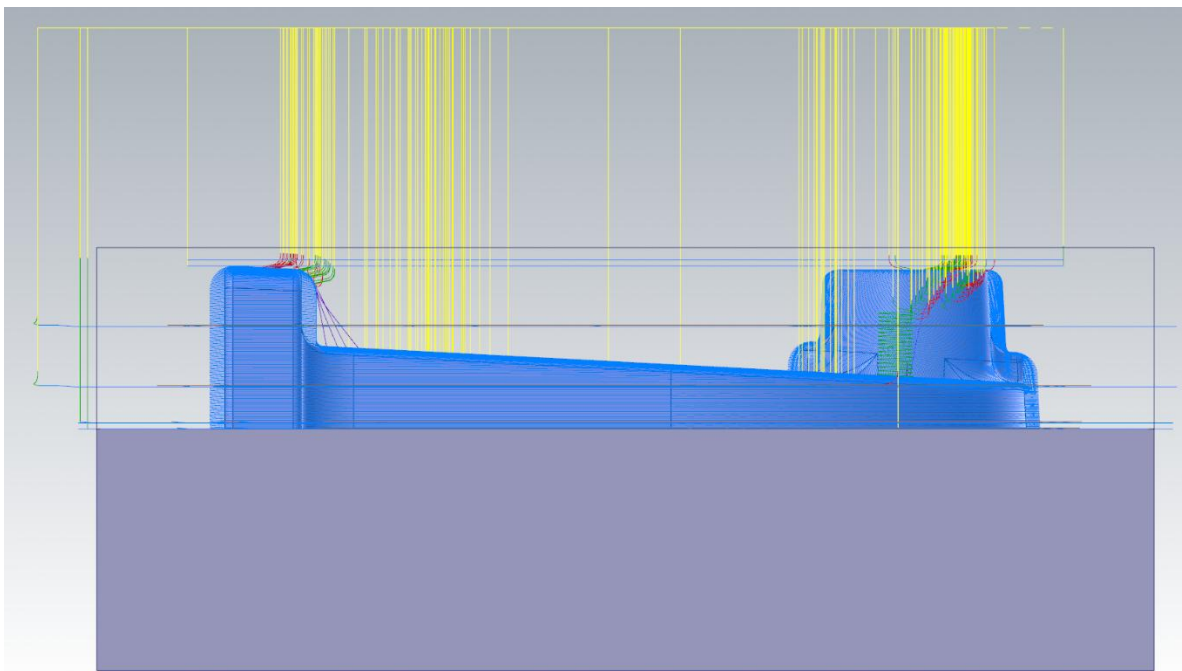


Figure 69: Side view of tool paths

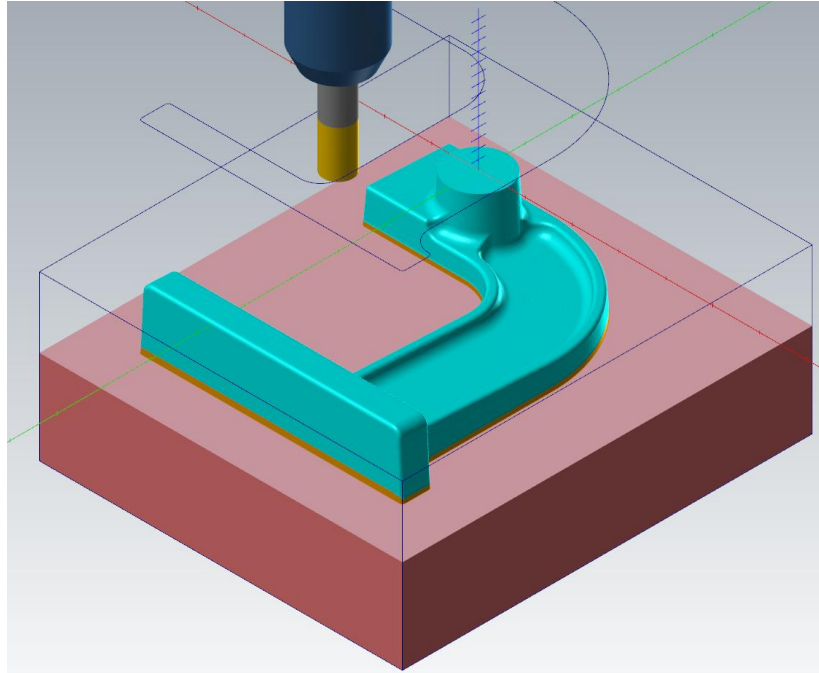


Figure 70: Full operation verification using color loop